SYNOPSIS

Title:

DevOps-Based Music Player Using Flutter, MySQL, and Python

Introduction:

The music industry has seen a rapid transformation with the advent of digital streaming platforms. This project aims to develop a music player application that allows users to stream, organize, and manage their music library efficiently. Built using Flutter for the frontend, Python for backend services, and MySQL for data storage, the application will leverage DevOps principles for continuous integration and deployment, ensuring seamless updates and scalability.

Scope of the Project:

- Development of a cross-platform music player with an intuitive user interface using Flutter.
- Integration with MySQL for secure music storage and metadata management.
- Backend services developed in Python to handle user authentication, music retrieval, and playlist management.
- Implementation of DevOps methodologies such as CI/CD pipelines for automated deployment and updates.
- Cloud-based storage solutions for scalability and performance optimization.

Objectives:

1. User-Friendly Interface: Design an interactive UI/UX for a smooth music playback experience.

2. Database Management: Store and manage music files, playlists, and user data in MySQL.

3. Efficient Backend Processing: Develop Python-based backend services to handle API requests.

4. DevOps Integration: Automate testing, deployment, and scaling using CI/CD pipelines.

5. Cross-Platform Compatibility: Ensure seamless operation on Android, iOS, and web platforms.

6. Security Measures: Implement authentication and encryption techniques to secure user data.

Resources Required:

1. Technologies & Tools: Flutter, Dart, Python, MySQL, Docker, Jenkins/GitHub Actions (for CI/CD), Firebase or AWS (for storage if needed).
2. Development Environment: VS Code/Android Studio, MySQL Server, Postman (for API testing).
3. Cloud Services: AWS/GCP/Azure for hosting and deployment.
4. Version Control: Git and GitHub/GitLab for code management.